FILE: C:\HCL_TRAINING\client\src\pages\ReportsPage.jsx

LINES: 378 | SIZE: 15.4 KB | LANGUAGE: JSX

MODULE: Recruitment Funnel Velocity, Skill Distribution, and CSV Export

```jsx
import React, { useState, useEffect } from 'react';
import { Link } from 'react-router-dom';
import {
  BarChart3,
  Download,
  Filter,
  Layers,
  Users,
  Briefcase,
  CheckCircle2,
  FileSpreadsheet,
  AlertCircle,
  RotateCcw,
  Sparkles,
} from 'lucide-react';
import {
  BarChart,
  Bar,
  XAxis,
  YAxis,
  CartesianGrid,
  Tooltip,
  ResponsiveContainer,
} from 'recharts';
import { api } from '../api/client';
import { ScoreBadge } from '../components/common/ScoreBadge';

export const ReportsPage = () => {
  const [data, setData] = useState(null);
  const [jobs, setJobs] = useState([]);
  const [selectedJob, setSelectedJob] = useState('all');
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);
  const [downloadingCSV, setDownloadingCSV] = useState(false);

  useEffect(() => {
    fetchJobs();
  }, []);

  useEffect(() => {
    fetchReports();
  }, [selectedJob]);

  const fetchJobs = async () => {
    try {
      const res = await api.get('/jobs');
      if (res.success) setJobs(res.data || []);
    } catch (e) {
      console.warn('Could not load jobs for dropdown filter:', e);
    }
  };

  const fetchReports = async () => {
    setLoading(true);
    setError(null);
    try {
      const query = selectedJob !== 'all' ? `?jobId=${selectedJob}` : '';
      const res = await api.get(`/reports/summary${query}`);
      if (res.success) {
        setData(res.data);
      } else {
        setError(res.message || 'Could not load recruitment analytics');
      }
    } catch (err) {
      console.error('Failed to fetch reports summary:', err);
      setError(err.message || 'Failed to connect to the reports API. Ensure backend is r
```

```
> unning.');
67 |     } finally {
68 |       setLoading(false);
69 |     }
70 |   };
71 |
72 |   const handleExportCSV = async () => {
73 |     setDownloadingCSV(true);
74 |     try {
75 |       const query = selectedJob !== 'all' ? `?jobId=${selectedJob}` : '';
76 |       const blob = await api.get(`/reports/export/csv${query}`);
77 |       const url = window.URL.createObjectURL(blob);
78 |       const a = document.createElement('a');
79 |       a.href = url;
80 |       a.download = `hiresense_rankings_${Date.now()}.csv`;
81 |       document.body.appendChild(a);
82 |       a.click();
83 |       a.remove();
84 |     } catch (err) {
85 |       alert('Failed to export CSV: ' + err.message);
86 |     } finally {
87 |       setDownloadingCSV(false);
88 |     }
89 |   };
90 |
91 |   const funnelData = data?.funnelData || [];
92 |   const skillDist = data?.skillDistribution || [];
93 |   const rankings = data?.candidateRankings || [];
94 |   const summaries = data?.jobSummaries || [];
95 |
96 |   return (
97 |     <div className="space-y-8 max-w-7xl mx-auto">
98 |       {/* Header */}
99 |       <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4 pb
> -2 border-b border-border">
100 |         <div>
101 |           <span className="text-xs font-semibold uppercase tracking-wider text-forest-70
> 0">
102 |             Talent Acquisition Analytics
103 |           </span>
104 |           <h1 className="font-editorial text-3xl font-bold text-ink mt-0.5">
105 |             Reports & Export Center
106 |           </h1>
107 |           <p className="text-xs text-ink-muted mt-1">
108 |             Funnel drop-off analytics, candidate rankings, and authorized CSV data expor
> t.
109 |           </p>
110 |         </div>
111 |
112 |         <div className="flex items-center gap-3">
113 |           <select
114 |             value={selectedJob}
115 |             onChange={(e) => setSelectedJob(e.target.value)}
116 |             className="bg-white border border-border rounded-lg text-xs font-medium text
> -ink px-3 py-2 focus:outline-none focus:border-forest-500 shadow-subtle"
117 |           >
118 |             <option value="all">All Job Openings</option>
119 |             {jobs.map((j) => (
120 |               <option key={j._id} value={j._id}>
121 |                 {j.title}
122 |               </option>
123 |             ))}
124 |           </select>
125 |
126 |           <button
127 |             onClick={handleExportCSV}
128 |             disabled={downloadingCSV}
129 |             className="inline-flex items-center gap-2 px-4 py-2 bg-forest-500 hover:bg-f
> orest-600 text-white rounded-lg text-xs font-semibold shadow-sm transition-colors disabl
> ed:opacity-50"
130 |           >
131 |             <FileSpreadsheet className="w-4 h-4" />
```

```
132 |             <span>{downloadingCSV ? 'Generating...' : 'Export Rankings CSV'}</span>
133 |           </button>
134 |         </div>
135 |       </div>
136 |
137 |       {/* Error Banner */}
138 |       {error && (
139 |         <div className="p-4 bg-red-50 border border-red-200 rounded-xl flex items-start
    >   justify-between gap-3 text-xs text-red-800">
140 |           <div className="flex items-start gap-2">
141 |             <AlertCircle className="w-4 h-4 text-red-600 mt-0.5 shrink-0" />
142 |             <div>
143 |               <p className="font-semibold text-red-900">Unable to load reports</p>
144 |               <p className="mt-0.5">{error}</p>
145 |             </div>
146 |           </div>
147 |           <button
148 |             onClick={fetchReports}
149 |             className="inline-flex items-center gap-1.5 px-3 py-1 bg-white border border
    >   -red-200 rounded-lg text-red-700 hover:bg-red-100 font-medium shrink-0 transition-colors
    >   "
150 |           >
151 |             <RotateCcw className="w-3.5 h-3.5" />
152 |             <span>Retry</span>
153 |           </button>
154 |         </div>
155 |       )}
156 |
157 |       {/* Loading Skeleton */}
158 |       {loading && !data && (
159 |         <div className="space-y-6 animate-pulse">
160 |           <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
161 |             <div className="h-72 bg-sage-50/60 rounded-2xl border border-border/60" />
162 |             <div className="h-72 bg-sage-50/60 rounded-2xl border border-border/60" />
163 |           </div>
164 |           <div className="h-64 bg-sage-50/60 rounded-2xl border border-border/60" />
165 |         </div>
166 |       )}
167 |
168 |       {/* Charts Section: Funnel & Skills Distribution */}
169 |       {(!loading || data) && (
170 |         <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
171 |           {/* Funnel Drop-off Bar Chart */}
172 |           <div className="surface-card rounded-2xl p-6 min-w-0">
173 |             <div className="flex items-center justify-between mb-1">
174 |               <h3 className="font-editorial text-lg font-semibold text-ink">
175 |                 Recruitment Funnel Velocity
176 |               </h3>
177 |               <span className="text-[11px] font-mono text-forest-700 bg-forest-50 px-2 p
    >   y-0.5 rounded-full font-medium">
178 |                 {funnelData.reduce((acc, curr) => acc + (curr.count || 0), 0)} Total App
    >   lications
179 |               </span>
180 |             </div>
181 |             <p className="text-xs text-ink-muted mb-4">
182 |               Candidate distribution and conversion attrition from New intake to Hired.
183 |             </p>
184 |
185 |             <div className="h-64 w-full min-h-[250px]">
186 |               <ResponsiveContainer width="100%" height="100%">
187 |                 <BarChart data={funnelData} margin={{ top: 10, right: 10, left: -20, bot
    >   tom: 0 }}>
188 |                   <CartesianGrid strokeDasharray="3 3" stroke="#EAECE5" vertical={false}
    >    />
189 |                   <XAxis dataKey="stage" stroke="#8C92A0" fontSize={11} tickLine={false}
    >    />
190 |                   <YAxis stroke="#8C92A0" fontSize={11} tickLine={false} allowDecimals={
    >   false} />
191 |                   <Tooltip
192 |                     contentStyle={{
193 |                       backgroundColor: '#FFFFFF',
194 |                       borderRadius: '8px',
```

```
195 |                       border: '1px solid #E2E4DE',
196 |                       fontSize: '12px',
197 |                     }}
198 |                   />
199 |                   <Bar dataKey="count" fill="#176B52" radius={[4, 4, 0, 0]} name="Candid
    >  ates" />
200 |                 </BarChart>
201 |               </ResponsiveContainer>
202 |             </div>
203 |           </div>
204 |
205 |           {/* Skills Pool Distribution */}
206 |           <div className="surface-card rounded-2xl p-6 min-w-0">
207 |             <div className="flex items-center justify-between mb-1">
208 |               <h3 className="font-editorial text-lg font-semibold text-ink">
209 |                 Skills Frequency Distribution
210 |               </h3>
211 |               <span className="text-[11px] font-mono text-ink-muted bg-sage-100 px-2 py-
    >  0.5 rounded-full font-medium">
212 |                 Top {skillDist.length} Competencies
213 |               </span>
214 |             </div>
215 |             <p className="text-xs text-ink-muted mb-4">
216 |               Top competencies available across screened candidates in the talent pool.
217 |             </p>
218 |
219 |             <div className="h-64 w-full min-h-[250px]">
220 |               {skillDist.length === 0 ? (
221 |                 <div className="h-full flex items-center justify-center text-xs text-ink
    >  -muted italic">
222 |                   No skills parsed yet
223 |                 </div>
224 |               ) : (
225 |                 <ResponsiveContainer width="100%" height="100%">
226 |                   <BarChart data={skillDist} layout="vertical" margin={{ top: 5, right:
    >  15, left: 10, bottom: 5 }}>
227 |                     <CartesianGrid strokeDasharray="3 3" stroke="#EAECE5" horizontal={fa
    >  lse} />
228 |                     <XAxis type="number" stroke="#8C92A0" fontSize={11} tickLine={false}
    >   allowDecimals={false} />
229 |                     <YAxis dataKey="skill" type="category" stroke="#8C92A0" fontSize={10
    >  } width={90} tickLine={false} />
230 |                     <Tooltip
231 |                       contentStyle={{
232 |                         backgroundColor: '#FFFFFF',
233 |                         borderRadius: '8px',
234 |                         border: '1px solid #E2E4DE',
235 |                         fontSize: '12px',
236 |                       }}
237 |                     />
238 |                     <Bar dataKey="count" fill="#90C7B4" radius={[0, 4, 4, 0]} name="Cand
    >  idates" />
239 |                   </BarChart>
240 |                 </ResponsiveContainer>
241 |               )}
242 |             </div>
243 |           </div>
244 |         </div>
245 |       )}
246 |
247 |       {/* Job Screening Summaries Table */}
248 |       {(!loading || data) && (
249 |         <div className="surface-card rounded-2xl p-6">
250 |           <h3 className="font-editorial text-lg font-semibold text-ink mb-1">
251 |             Role-Wise Screening Summaries
252 |           </h3>
253 |           <p className="text-xs text-ink-muted mb-4">
254 |             Aggregate performance and average score benchmarks across open positions.
255 |           </p>
256 |
257 |           <div className="overflow-x-auto">
258 |             <table className="w-full text-left text-xs">
```

```
259 |               <thead className="bg-[#FAFBF9] border-b border-border text-ink-muted upper
    >  case font-semibold text-[10px] tracking-wider">
260 |                 <tr>
261 |                   <th className="py-3 px-4">Role Title</th>
262 |                   <th className="py-3 px-4">Department</th>
263 |                   <th className="py-3 px-4">Total Screened</th>
264 |                   <th className="py-3 px-4">Shortlisted</th>
265 |                   <th className="py-3 px-4">Hired</th>
266 |                   <th className="py-3 px-4">Avg Match Score</th>
267 |                 </tr>
268 |               </thead>
269 |               <tbody className="divide-y divide-border font-medium">
270 |                 {summaries.length === 0 ? (
271 |                   <tr>
272 |                     <td colSpan="6" className="py-8 text-center text-ink-muted">
273 |                       No active job openings found
274 |                     </td>
275 |                   </tr>
276 |                 ) : (
277 |                   summaries.map((s) => (
278 |                     <tr key={s.jobId} className="hover:bg-sage-50/40 transition-colors">
279 |                       <td className="py-3 px-4 font-bold text-ink">{s.title}</td>
280 |                       <td className="py-3 px-4 text-ink-muted">{s.department}</td>
281 |                       <td className="py-3 px-4 font-mono">{s.totalCandidates}</td>
282 |                       <td className="py-3 px-4 font-mono text-forest-700">{s.shortlisted
    >  Candidates}</td>
283 |                       <td className="py-3 px-4 font-mono font-bold text-forest-800">{s.h
    >  iredCandidates}</td>
284 |                       <td className="py-3 px-4 font-mono">
285 |                         <ScoreBadge score={s.avgMatchScore} size="sm" />
286 |                       </td>
287 |                     </tr>
288 |                   ))
289 |                 )}
290 |               </tbody>
291 |             </table>
292 |           </div>
293 |         </div>
294 |       )}
295 |
296 |       {/* Ranked Candidate Leaderboard */}
297 |       {(!loading || data) && (
298 |         <div className="surface-card rounded-2xl p-6">
299 |           <div className="flex items-center justify-between mb-4">
300 |             <div>
301 |               <h3 className="font-editorial text-lg font-semibold text-ink">
302 |                 Ranked Candidate Roster ({rankings.length})
303 |               </h3>
304 |               <p className="text-xs text-ink-muted">
305 |                 Top scoring candidate evaluations sorted strictly by evidence-backed sco
    >  res.
306 |               </p>
307 |             </div>
308 |           </div>
309 |
310 |           <div className="overflow-x-auto">
311 |             <table className="w-full text-left text-xs">
312 |               <thead className="bg-[#FAFBF9] border-b border-border text-ink-muted upper
    >  case font-semibold text-[10px] tracking-wider">
313 |                 <tr>
314 |                   <th className="py-3 px-4">Rank</th>
315 |                   <th className="py-3 px-4">Candidate</th>
316 |                   <th className="py-3 px-4">Target Job</th>
317 |                   <th className="py-3 px-4">Overall Score</th>
318 |                   <th className="py-3 px-4">Skill Align</th>
319 |                   <th className="py-3 px-4">Semantic</th>
320 |                   <th className="py-3 px-4">Exp</th>
321 |                   <th className="py-3 px-4 text-right">Profile</th>
322 |                 </tr>
323 |               </thead>
324 |               <tbody className="divide-y divide-border">
325 |                 {rankings.length === 0 ? (
```

```
326 |                   <tr>
327 |                     <td colSpan="8" className="py-8 text-center text-ink-muted">
328 |                       No candidate screening records available yet. Upload resumes to ge
    >   nerate rankings.
329 |                     </td>
330 |                   </tr>
331 |                 ) : (
332 |                   rankings.map((r, idx) => (
333 |                     <tr key={r._id} className="hover:bg-sage-50/40 transition-colors">
334 |                       <td className="py-3 px-4 font-mono font-bold text-ink-muted">
335 |                         #{idx + 1}
336 |                       </td>
337 |                       <td className="py-3 px-4">
338 |                         <Link
339 |                           to={`/candidates/${r.candidateId?._id || ''}`}
340 |                           className="font-bold text-ink hover:text-forest-700"
341 |                         >
342 |                           {r.candidateId?.name || 'Evaluated Candidate'}
343 |                         </Link>
344 |                         <span className="text-[10px] text-ink-subtle block">
345 |                           {r.candidateId?.email || 'Confidential'}
346 |                         </span>
347 |                       </td>
348 |                       <td className="py-3 px-4 text-ink-muted">{r.jobId?.title || 'Open
    >   Role'}</td>
349 |                       <td className="py-3 px-4">
350 |                         <ScoreBadge score={r.overallScore} size="sm" />
351 |                       </td>
352 |                       <td className="py-3 px-4 font-mono">{r.componentScores?.skillAlign
    >   ment ?? 0}%</td>
353 |                       <td className="py-3 px-4 font-mono">{r.componentScores?.semanticSi
    >   milarity ?? 0}%</td>
354 |                       <td className="py-3 px-4 font-mono">{r.candidateId?.totalExperienc
    >   eYears ?? 0}y</td>
355 |                       <td className="py-3 px-4 text-right">
356 |                         {r.candidateId?._id ? (
357 |                           <Link
358 |                             to={`/candidates/${r.candidateId._id}`}
359 |                             className="text-forest-600 hover:text-forest-800 font-semibo
    >   ld"
360 |                           >
361 |                             Review &rarr;
362 |                           </Link>
363 |                         ) : (
364 |                           <span className="text-ink-subtle text-[11px]">Archived</span>
365 |                         )}
366 |                       </td>
367 |                     </tr>
368 |                   ))
369 |                 )}
370 |               </tbody>
371 |             </table>
372 |           </div>
373 |         </div>
374 |       )}
375 |     </div>
376 |   );
377 | };
378 |
```